

STA399Y5Y - ROP

Enhancing Bayesian Analyses: Adjusting Improper Priors with Cross-Validation Techniques

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Objective

- The objective of this study is to explore a methodology aimed at refining Bayesian analyses when confronted with an improper prior $\pi(\theta)$ and a set of data $y_1, \dots, y_n$.
- The methodology involves partitioning the dataset into two distinct subsamples: $y_1, \dots, y_l$ and $y_1^*, \dots, y_m^*$, where the total number of observations is denoted by $n = l + m$.
- This process unfolds in two stages:
 - ➊ In the initial stage, we utilize the improper prior $\pi(\theta)$ along with the first subsample $y_1, \dots, y_l$ to derive the posterior distribution $\pi(\theta|y_1, \dots, y_l)$.
 - ➋ Subsequently, in the second stage, we leverage this computed posterior as the new prior, denoted as $\pi^*(\theta)$, and apply it to the second subsample $y_1^*, \dots, y_m^*$ to obtain $\pi^*(\theta|y_1^*, \dots, y_m^*)$.

Objective, continue ...

Several key questions naturally arise from this methodology:

- ➊ How do we effectively partition the dataset into the two subsamples?
- ➋ What is the efficacy of this approach in improving Bayesian analyses?
- ➌ What are the potential real-world applications of this refined methodology?

Introduction

- In any inference problem, we start with a statistical model:

$$\{f_{\theta}(\cdot) = f(\cdot|\theta) : \theta \in \Theta\}.$$

- The statistical model describes the statistician's uncertainty about how the observed data $y_1, \dots, y_n$ were produced.
- Traditional statistical approaches rely solely on the model and the observed data to draw inferences about the unknown parameter θ .
- If we accept the principle that

all uncertainties must be quantified with probabilities,

then relying solely on a model is inadequate. This approach fails to provide a method for making probability-based assertions concerning the unknown true value of θ .

Introduction, continue ...

- We complete the description by prescribe a probability distribution for θ .
- This is the **Bayesian approach** to inference.
- So, Bayesian inference differs from classical inference in that it specifies a probability distribution for the parameter(s) of interest.
- Thus, the **statistical model in Bayesian context** is written as:

$$y_1, \dots, y_n \stackrel{iid}{\sim} f(\cdots | \theta), \quad \theta \sim \pi(\theta).$$

- iid: independent and identically distributed.

Why Use Bayesian Inference? [Some reasons]

- 1 Incorporate previous knowledge we have about a parameter of interest (**prior distribution**).
- 2 **Bayes' theorem:** logically update our knowledge about the parameter after observing sample data (**posterior distribution**).
- 3 A clear (common-sense) interpretation of statistical conclusions.
 - **Example:** Bayesian (probability) interval for θ can be directly regarded as the probability of containing θ .
- 4 Flexibility and generality allow it to cope with complex problems.
- 5 Many subjects such as data mining and machine learning embrace Bayesian methods very strongly.

Why Do Statisticians Use Classical (Standard, Frequentist) Methods?

- 1 If there is no prior information available about the parameter(s).
- 2 If the parameter(s) of interest is/are truly fixed (without the possibility of changing) [as is possible in a highly controlled experiment].
- 3 Many statisticians prefer to develop statistical theory without the additional ingredients necessary for a full probability description of the unknown.
- 4 Historically, Bayesian analyses had been infeasible due to a lack of computational algorithms.

Bayesian Versus Frequentist Methods

- **Frequentist:**

- Parameters are constant to be estimate.
- CLT.
- Point estimation and interval estimation.
- model: $y_1, \dots, y_n \stackrel{iid}{\sim} f(\cdot|\theta)$.

- **Bayesian:**

- Parameters are random (have a distribution).
- Prior and Posterior.
- all information are contained in posterior distribution.
- Model: $y_1, \dots, y_n \stackrel{iid}{\sim} f(\cdot|\theta), \theta \sim \pi(\theta)$.

Three steps of Bayesian data analysis

- Bayesian data analysis is carried out into the following three steps:
 - ① Choose the **prior distribution** $\pi(\theta)$. " This reflects our beliefs about the true value of the parameter θ .
 - ② Choose the statistical model $f(\cdot|\theta)$. This reflect our beliefs about y given θ .
 - ③ After observing that data $y = (y_1, \dots, y_n)$, we update our beliefs by calculating the **posterior distribution** $f(\theta|y)$.
- **Always remember:** The **posterior distribution** of θ is the conditional distribution of θ given the data y .

How to Select a Prior?

- A **natural question to ask is:** *Where do these beliefs come from in an application?*
- An easy answer: they come from previous experience.
- To be honest: this is rarely the case.
- One has to admit that the prior, as well as the statistical model, is often a somewhat arbitrary construction used to drive the statistician's investigations.

How to Select a Prior?, continue ...

- This raises the issue as to whether or not the inferences derived have any relevance to the practical context, if the model ingredients suffer from this arbitrariness.
- This is where the concept of **model checking** and **prior-data conflict** comes into play, a topic we will discuss later.
- At this point, we will assume that all the ingredients make sense, but remember that in an application, these must be checked if the inferences taken are to be practically meaningful.

Notations

- We use the terms 'distribution' and 'density' interchangeably.
- The same notation is used for continuous density functions and discrete probability mass functions.
- $\Pr$ or P : denote probability.
- When using a standard distribution, we use a notation based on the name of the distribution.
 - **Example:** If θ has a normal distribution with mean μ and variance σ^2 , we write:
 - $\theta \sim N(\mu, \sigma^2)$ or
 - $\pi(\theta) = N(\theta|\mu, \sigma^2)$ or,
 - $\pi(\theta|\mu, \sigma^2) = N(\theta|\mu, \sigma^2)$.

Exchangeability

- The usual starting point of a statistical analysis is the assumption that the data values $y_1, y_2, \dots, y_n$ in the sample are exchangeable.
- That is, reordering the data values does not change the model.
- Mathematically, for any permutation π of $\{1, 2, \dots, n\}$ we have

$$f(y_1, \dots, y_n) = f(y_{\pi_1}, \dots, y_{\pi_n}).$$

- We commonly model data from an exchangeable distribution as independently and identically distributed (iid) given some unknown parameter vector θ .

Exchangeability and iid

Theorem

If $\theta \sim \pi(\theta)$ and $y_1, \dots, y_n | \theta$ are iid, then the $y_1, \dots, y_n$ are exchangeable. [That is, marginally (unconditionally on θ), the data are exchangeable.]

Proof:

Exchangeability

- From the previous theorem:

$y_1, \dots, y_n | \theta$ are iid and $\theta \sim \pi(\theta) \Rightarrow y_1, \dots, y_n$ are exchangeable.

- What about an arrow in the other direction (i.e. the converse)?
- A famous theorem (de Finetti's Theorem) shows that

$y_1, \dots, y_n | \theta$ are iid and $\theta \sim \pi(\theta) \Leftrightarrow y_1, \dots, y_n$ are exchangeable for all n .

- When is the condition $y_1, \dots, y_n$ are exchangeable for all n reasonable?
- Examples (A First Course in Bayesian Statistical Methods by Peter D. Hoff):
 - $y_1, \dots, y_n$ are outcomes of a repeatable experiment;
 - $y_1, \dots, y_n$ are sampled from a finite population with replacement;
 - $y_1, \dots, y_n$ are sampled from an infinite population without replacement

Notation

- **Note:** In statistics course, upper-case letters (such as $X, Y, U, V, \dots$) denote random variables and lower-case letters (like $x, y, u, v, \dots$) represent observed (fixed) values. In this course, by convention, lower-case letters (such that $x, y, u, v, \dots$) are used to denote random variables.
- The space of all possible values of the r.v. is called the *support*.

Single-Parameter model - Binomial

- Definition of Binomial Distribution: Binomial Distribution $\text{Bin}(n, \theta)$ is a sequence of n exchangeable "Bernoulli Trials".

data point $y_1, \dots, y_n$ each of which is either 0 or 1 .

- Our Binomial Model will be:

$$\theta = \frac{\sum_{i=1}^N y_i}{N}, \text{ for } \theta \in [0, 1]$$

$$f(y_1, \dots, y_n | \theta) = \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{n - \sum_{i=1}^n y_i}$$

- Because of exchangeability, the data can be summarized by the total number of successes in the n trials, which we denote here by y .

$$f(Y = y | \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

Property of Binomial model

- **Importance of Inference for exchangeable binary data:** $\sum_{i=1}^n y_i$ is a *sufficient statistic* for θ and $f(y_1, \dots, y_n | \theta)$, in this case the *sufficient statistic* has a Binomial distribution with parameters (n, θ) .
- Property of Binomial Model:

$$\text{Mean: } E[X] = n\theta$$

$$\text{Variance: } \text{Var}[X] = n\theta(1 - \theta)$$

Single-Parameter model - Beta

- Now, continue from our Binomial Model, since we have already selected our statistical model we now have to select a prior distribution $\pi(\theta)$ to find our posterior distribution. However, before we select our prior distribution we need to take a look at Beta Distribution.
- The pdf of Beta Distribution $(\alpha, \beta) = \int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta$
for $\theta \in [0, 1]$
- Note: $\Gamma(\alpha) = (\alpha - 1)!$

Property of Beta model

- Property of Beta Model:

$$\text{Mean: } E[X] = \frac{\alpha}{\alpha + \beta}$$

$$\text{Variance: } Var[X] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

- Shape of the probability density function of the Beta model with different values in parameter Beta(α, β):

$$\alpha = \beta = 0.5$$

$$\alpha = 0.5, \beta = 1$$

$$\alpha = 1, \beta = 0.5$$

$$\alpha = \beta = 1$$

$$\alpha = 10, \beta = 10$$

Application of Bayes' rule and posterior density model for θ

- Choose the **prior distribution**: Continuous uniform distribution for $\theta, \in [0, 1]$. $\pi(\theta) \sim \text{unif}(0, 1) \sim \text{Beta}(1, 1)$
- Use the chosen **statistical model(likelihood)**: Binomial model
 $f(y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$.
- Use **Bayes' rule** to obtain **posterior distribution**: $f(\theta|y) = \frac{\pi(\theta)f(y|\theta)}{\pi(y)}$.

$$f(\theta|y) \propto \pi(\theta)f(y|\theta)$$

$$f(\theta|y) \propto \theta^{(y+1)-1} (1 - \theta)^{(n-y+1)-1}$$

$$f(\theta|y) \sim \text{Beta}(y + 1, n - y + 1)$$

- Note:** $\pi(y)$ can be calculated numerically, in the discrete case $\pi(y) = \sum_{\theta} \pi(\theta)f(y|\theta)$ or in the continuous case $\pi(y) = \int_{\theta} \pi(\theta)f(y|\theta)d\theta$.

Conjugacy of Beta prior

Theorem

The beta prior distribution is a conjugate family for the binomial likelihood. This can be extended to Bernoulli and Geometric likelihood as well.

Proof: Show that a beta prior distribution and a binomial statistical model lead to a beta posterior distribution.

Special Rule about Conjugate family

Theorem

Conjugate priors only exist when our statistical model comes from the exponential family.

Exponential family distributions: Normal, Gamma, Beta, Bernoulli, Poisson, Exponential, Chi-square, Geometric, Binomial, and Negative Binomial(with a fixed number of trials/failures) etc.

Bayesian Inference

- We already know that with a Beta prior distribution $\pi(\theta) \sim \text{Beta}(\alpha, \beta)$, and a Binomial statistical model(likelihood) $f(y|\theta) \sim \text{Binomial}(n, y)$. We can obtain a Beta posterior distribution $f(\theta|y) \sim \text{Beta}(\alpha + y, \beta + n - y)$.
- **Posterior Expectation:** $E[\theta|y] = \frac{\alpha+y}{\alpha+\beta+n} = \frac{\alpha+\beta}{\alpha+\beta+n} \frac{\alpha}{\alpha+\beta} + \frac{n}{\alpha+\beta+n} \frac{y}{n}$
- **Posterior Variance:**
$$V[\theta|y] = \frac{(\alpha+y)(\beta+n-y)}{(\alpha+\beta+n)^2(\alpha+\beta+n+1)}$$
$$= \frac{\alpha+y}{\alpha+\beta+n} \frac{\beta+n-y}{\alpha+\beta+n} \frac{1}{\alpha+\beta+n+1} = \frac{E[\theta|y][1-E[\theta|y]]}{\alpha+\beta+n+1}$$
- With a large sample size of n , we will see posterior Expectation and posterior variance converge.
- Credible Interval and Normal approximation of posterior distribution

Posterior predictive probability

Beta posterior distribution $\text{Beta}(\alpha^* = \alpha + y, \beta^* = \beta + n - y)$.

- $p(\tilde{y} = 1|y) = \int_0^1 p(\tilde{y} = 1, \theta|y)d\theta = \int_0^1 \theta p(\theta|y)d\theta = E[\theta|y]$.
- $p(\tilde{y} = 0|y) = \int_0^1 p(\tilde{y} = 0, \theta|y)d\theta = \int_0^1 (1 - \theta)p(\theta|y)d\theta = 1 - E[\theta|y]$.

Posterior predictive probability of $\tilde{y}$ estimated success with a future sample size n^* :

- $p(\tilde{y}|y) = \int_0^1 \binom{n^*}{\tilde{y}} \theta^{\tilde{y}} (1 - \theta)^{n^* - \tilde{y}} \frac{\Gamma(\alpha^* + \beta^*)}{\Gamma(\alpha^*)\Gamma(\beta^*)} \theta^{\alpha^* - 1} (1 - \theta)^{\beta^* - 1} d\theta$.
- $p(\tilde{y}|y) = \binom{n^*}{\tilde{y}} \frac{\Gamma(\alpha^*)\Gamma(\beta^*)}{\Gamma(\alpha^* + \beta^*)} \frac{\Gamma(\alpha^* + \tilde{y})\Gamma(n^* + \beta^* - \tilde{y})}{\Gamma(\alpha^* + \beta^* + n^*)}$

Poisson & Gamma model

$$P(y^*|y) = \int_0^1 P(y^*|\pi)P(\pi|y) d\pi \quad \text{and} \quad P(y^*|x) = \frac{P(y^* \cap y)}{P(y)}$$

Thus

$$\begin{aligned} P(y^*|y) &= \int_0^1 P(y^*|\pi) \left[\frac{P(y|\pi)P(\pi)}{\int_0^1 P(y|\pi)P(\pi) d\pi} \right] d\pi \\ &= \frac{\int_0^1 P(y^*|\pi)P(y|\pi)P(\pi) d\pi}{\int_0^1 P(y|\pi)P(\pi) d\pi} \\ &= \frac{\int_0^1 P(y^*, y|\pi)P(\pi) d\pi}{\int_0^1 P(y, \pi) d\pi} \\ &= \frac{\int_0^1 P(y^*, y, \pi) d\pi}{P(y)} \\ &= \frac{P(y^*, y)}{P(y)} \end{aligned}$$

Poisson & Gamma model

Parameter: μ (mean)

Likelihood: $\text{Poisson}(\mu)$

$$P(y|\mu) = \frac{\mu^y}{y!} e^{-\mu} \quad y = 0, 1, 2, 3, \dots$$

Prior: $\text{Gamma}(r, v)$

$$P(\mu) = \frac{v^r}{\Gamma(r)} \mu^{r-1} e^{-\mu v}$$

Find $P(\mu|y)$

$$P(\mu|y) = \frac{P(y|\mu)P(\mu)}{\int_0^\infty P(y|\mu)P(\mu)d\mu}$$

Denominator

$$\int_0^\infty P(y|\mu)P(\mu)d\mu = \int_0^\infty \frac{\mu^y}{y!} e^{-\mu} \frac{v^r}{\Gamma(r)} \mu^{r-1} e^{-\mu v} d\mu$$

Poisson & Gamma model

$$\begin{aligned}
 &= \frac{v^r}{y! \Gamma(r)} \int_0^\infty \mu^{y+r-1} e^{-\mu-\mu v} d\mu \\
 &= \frac{v^r}{y! \Gamma(r)} \int_0^\infty \mu^{(y+r)-1} e^{-\mu(1+v)} d\mu
 \end{aligned}$$

Let $r^* = y + r$, $v^* = 1 + v$,

$$\begin{aligned}
 &= \frac{v^r}{y! \Gamma(r)} \frac{\Gamma(y+r)}{(v+1)^{y+r}} \int_0^\infty \frac{(v+1)^{y+r}}{\Gamma(y+r)} \mu^{(y+r)-1} e^{-\mu(1+v)} d\mu \\
 &= \frac{v^r}{y! \Gamma(r)} \frac{\Gamma(y+r)}{(v+1)^{y+r}}
 \end{aligned}$$

Finally

$$\begin{aligned}
 P(\mu|y) &= \frac{\frac{\mu^y}{y!} e^{-\mu} \frac{v^r}{\Gamma(r)} \mu^{r-1} e^{-\mu v}}{\frac{v^r}{y! \Gamma(r)} \frac{\Gamma(y+r)}{(v+1)^{y+r}}} \\
 &= \frac{(v+1)^{(y+r)}}{\Gamma(y+r)} \mu^{(y+r)-1} e^{-(v+1)\mu} \sim \text{Gamma}(r^* = y + r, v^* = 1 + v)
 \end{aligned}$$

Poisson & Gamma model

Parameter: μ

Prior: $\text{Gamma}(r, v) = P(\mu)$

Likelihood: $P(y_1, y_2, \dots, y_n | \mu)$

Find $P(\mu | y_1, y_2, \dots, y_n)$

Given the likelihood:

$$\begin{aligned}
 P(y_1, \dots, y_n | \mu) &= \prod_{i=1}^n P(y_i | \mu) \\
 &= \left(\frac{\mu^{y_1} e^{-\mu}}{y_1!} \right) \times \dots \times \left(\frac{\mu^{y_n} e^{-\mu}}{y_n!} \right) \\
 &= \frac{\mu^{y_1 + \dots + y_n} e^{-n\mu}}{y_1! y_2! \dots y_n!} \\
 &= \frac{\mu^{\sum_{i=1}^n y_i} e^{-n\mu}}{\prod_{i=1}^n y_i!}
 \end{aligned}$$

Poisson & Gamma model

Denominator

$$\begin{aligned} \int_0^\infty \frac{1}{y_1! \cdots y_n!} \mu^{\sum_{i=1}^n y_i} e^{-n\mu} \frac{v^r}{\Gamma(r)} \mu^{r-1} e^{-v\mu} d\mu \\ = \frac{1}{y_1! \cdots y_n!} \frac{v^r}{\Gamma(r)} \int_0^\infty \mu^{(\sum_{i=1}^n y_i + r) - 1} e^{-\mu(n+v)} d\mu \end{aligned}$$

Doing something very similar to what we did in the single observation case, we have:

$$\begin{aligned} P(\mu|y_1, \dots, y_n) &= \frac{P(y_1, \dots, y_n|\mu)P(\mu)}{\int_0^\infty P(y_1, \dots, y_n|\mu)P(\mu) d\mu} \\ &\sim \text{Gamma} \left(r^* = \sum_{i=1}^n y_i + r, v^* = n + v \right) \end{aligned}$$

Poisson & Gamma model

Obtaining Updating Rules Using Shortcut:

$$\begin{aligned}P(\mu|y_1, \dots, y_n) &\propto P(y_1, \dots, y_n|\mu)P(\mu) \\&\propto \mu^{\sum_{i=1}^n y_i} e^{-n\mu} \mu^{r-1} e^{-v\mu} \\&\propto \mu^{(\sum_{i=1}^n y_i + r) - 1} e^{-\mu(n+v)}\end{aligned}$$

Poisson & Gamma model

Non-informative Prior for μ

Definition of Fisher Information:

$$I(\theta) = \mathbb{E} \left[-\frac{\partial^2 \log P(X|\theta)}{\partial^2 \theta^2} \right]$$

Let x be a random variable with probability density function $P(x|\theta)$. The expected Fisher Information measure of θ through x is defined by the above expression.

Poisson & Gamma model

Definition

Consider an observation x with PDF function $P(x|\theta)$. The Jeffrey's non-informative prior has density given by

$$P(\theta) \propto [I(\theta)]^{1/2}$$

Example Let $X = (x_1, x_2, \dots, x_n)$ be a sample of $\text{Poisson}(\theta)$ variables.

$$P(x_1, x_2, \dots, x_n | \theta) = \frac{1}{x_1! \cdots x_n!} \theta^{\sum_{i=1}^n x_i} e^{-n\theta}$$

$$\ln(P((x_1, x_2, \dots, x_n | \theta))) = \ln(x_1, x_2, \dots, x_n) + \sum_{i=1}^n x_i \ln(\theta) - n\theta$$

The first derivative of the log likelihood with respect to θ is:

$$\frac{\partial}{\partial \theta} \ln P(x_1, \dots, x_n | \theta) = \frac{\sum_{i=1}^n x_i}{\theta} - n = \sum_{i=1}^n x_i (\theta)^{-1} - n$$

Poisson & Gamma model

The second derivative of the log likelihood is:

$$\frac{\partial^2}{\partial \theta^2} \ln P(x_1, \dots, x_n | \theta) = -\frac{\sum_{i=1}^n x_i}{\theta^2}$$

Thus, the Fisher Information $I(\theta)$ is:

$$\begin{aligned} I(\theta) &= -\mathbb{E} \left[-\frac{\sum_{i=1}^n x_i}{\theta^2} \right] \\ &= \frac{1}{\theta^2} \sum_{i=1}^n \mathbb{E}(x_i) \\ &= \frac{1}{\theta^2} \times n\theta \\ &= \frac{n}{\theta} \end{aligned}$$

Finally, $P(\theta) \propto \frac{1}{\theta} \propto \theta^{-1/2}$

Poisson & Gamma model

Let Y be the number of accidents occurring in an industrial plant, Y is described by a Poisson process with mean accidents every three months. Suppose that 5 possible values of μ are $1/3$, $2/3$, 1 , $4/3$, and $5/3$. We do not have any reason to give any possible value more weight than any other value, so we give them equal prior weight. During the last three months, NO accidents occur.

Answer the following questions. You have to show **all your work** to get full credit.

a) Find the posterior distribution, i.e., $p(\mu|data)$.

$$P(y|\pi) = \frac{e^{-\mu}}{y!} \mu^y$$

$$P(y=0|\mu=1/3) = \frac{(1/3)^0}{0!} e^{-1/3} = e^{-1/3}$$

Poisson & Gamma model

Possible Value π_j	Prior $P(\pi_j)$	Likelihood $P(y \pi_j)$	Prior $\times$ Likelihood $P(y \pi_j) \times P(\pi_j)$	Posterior
1/3	0.20	0.7165	0.1433	0.3494
2/3	0.20	0.5134	0.1027	0.2504
1	0.20	0.3679	0.0736	0.1795
4/3	0.20	0.2636	0.0527	0.1285
5/3	0.20	0.1889	0.0378	0.0922

Poisson & Gamma model

b) Find the posterior mean, i.e., $E(\mu|data)$.

$$\begin{aligned} E(\mu|data) &= \sum_{\text{all } \mu's} \mu P(\mu|data) \\ &= \left(\frac{1}{3} \times 0.3494\right) + \cdots + \left(\frac{5}{3} \times 0.0922\right) \\ &\approx 0.7879 \end{aligned}$$

Poisson & Gamma model

c) Find $p(\mu \leq 1|data)$.

$$\begin{aligned}P(\mu \leq 1|data) &= 0.3494 + 0.2504 + 0.1795 \\ &\approx 0.7793\end{aligned}$$

Likelihood of Single Observation

The conditional observation distribution of $y \mid \mu$ is Normal with mean μ and variance σ^2 , which is **known**. Its density is given by:

$$f(y \mid \mu) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right)$$

The part that doesn't depend on the parameter μ can be absorbed into the proportionality constant. Thus, the likelihood shape is given by:

$$f(y \mid \mu) \propto \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right)$$

where y is held constant at the observed value and μ is allowed to vary over all possible values.

Likelihood of Single Observation

Usually, we have a random sample $Y_1, Y_2, \dots, Y_n$ of observations instead of a single observation. The observations in a random sample are all independent of each other, so the joint likelihood of the sample is the product of the individual observation likelihoods. This gives:

$$f(y_1, \dots, y_n \mid \mu) = f(y_1 \mid \mu) \times f(y_2 \mid \mu) \times \dots \times f(y_n \mid \mu).$$

We are considering the case where the distribution of each observation $y_i \mid \mu$ is Normal with mean μ and variance σ^2 , which is **known**.

Finding the posterior probabilities analyzing the sample all at once

Each observation is Normal, so it has a Normal likelihood. This gives the joint likelihood:

$$f(y_1, \dots, y_n \mid \mu) \propto e^{-\frac{1}{2\sigma^2}(y_1 - \mu)^2} \times e^{-\frac{1}{2\sigma^2}(y_2 - \mu)^2} \times \dots \times e^{-\frac{1}{2\sigma^2}(y_n - \mu)^2}$$

After "a little bit" of algebra we get:

$$f(y_1, \dots, y_n \mid \mu) \propto e^{-\frac{n}{2\sigma^2}(\mu^2 - 2\mu\bar{y} + \bar{y}^2)} \times e^{-\frac{n}{2\sigma^2}(\frac{1}{n} \sum_{i=1}^n y_i^2 - \bar{y}^2)}$$

When we absorb the part that doesn't involve μ into the proportionality constant we get:

$$f(y_1, \dots, y_n \mid \mu) \propto e^{-\frac{1}{2\frac{\sigma^2}{n}}(\bar{y} - \mu)^2}$$

We recognize that this likelihood has the shape of a Normal distribution with mean μ and variance $\frac{\sigma^2}{n}$. So the joint likelihood of the random sample is proportional to the likelihood of the sample mean, which is:

$$f(\bar{y} \mid \mu) \propto e^{-\frac{1}{2\frac{\sigma^2}{n}}(\bar{y} - \mu)^2}$$

Flat Prior Density for μ

The flat prior gives each possible value of μ equal weight. It does not favor any value over any other, hence $g(\mu) = 1$. The flat prior is not really a proper prior distribution since $-\infty < \mu < \infty$, so it can't integrate to 1. Nevertheless, this **improper** prior works out all right. Even though the prior is improper, the posterior will integrate to 1, so it is proper.

A single Normal observation y

Let y be a Normally distributed observation with mean μ and known variance σ^2 . The likelihood is:

$$f(y \mid \mu) \propto \exp \left(-\frac{1}{2\sigma^2} (y - \mu)^2 \right),$$

if we ignore the constant of proportionality.

Since the prior always equals 1, the posterior is proportional to this. Rewrite it as:

$$g(\mu \mid y) \propto \exp \left(-\frac{1}{2\sigma^2} (y - \mu)^2 \right).$$

We recognize from this shape that the posterior is a Normal distribution with mean y and variance σ^2 .

Normal Prior Density for μ

The observation y is a random variable taken from a normal distribution with mean μ and variance σ^2 which is assumed **known**. We have a prior distribution that is normal with mean m and variance s^2 . The shape of the prior density is given by:

$$g(\mu) \propto \exp\left(-\frac{1}{2s^2}(\mu - m)^2\right).$$

Posterior

The prior times the likelihood:

$$g(\mu) \times f(y|\mu) \propto \exp \left(-\frac{1}{2} \left[\frac{(\mu - m)^2}{s^2} + \frac{(y - \mu)^2}{\sigma^2} \right] \right).$$

After simplifying:

$$g(\mu) \times f(y|\mu) \propto \exp \left(-\frac{1}{\frac{2\sigma^2 s^2}{\sigma^2 + s^2}} \left[\mu - \frac{\sigma^2 m + s^2 y}{\sigma^2 + s^2} \right]^2 \right).$$

Posterior distribution: We recognize from this shape that the posterior is a Normal distribution having mean and variance given by

$$m' = \frac{\sigma^2 m + s^2 y}{\sigma^2 + s^2}, \quad (s')^2 = \frac{\sigma^2 s^2}{\sigma^2 + s^2}.$$

Simple updating rule for Normal family

First, we introduce the precision of a distribution, which is the reciprocal of the variance. For a posterior distribution with variance s'^2 , the precision is:

$$\frac{1}{(s')^2} = \left(\frac{\sigma^2 s^2}{\sigma^2 + s^2} \right)^{-1} = \left(\frac{\sigma^2 + s^2}{\sigma^2 s^2} \right) = \frac{1}{s^2} + \frac{1}{\sigma^2}.$$

Thus the posterior precision equals prior precision plus the observation precision.

Simple updating rule for Normal family

The posterior mean is given by:

$$m' = \frac{\sigma^2 m + s^2 y}{\sigma^2 + s^2} = \frac{\sigma^2}{\sigma^2 + s^2} \times m + \frac{s^2}{\sigma^2 + s^2} \times y$$

This can be simplified to

$$m' = \frac{1/s^2}{1/\sigma^2 + 1/s^2} \times m + \frac{1/\sigma^2}{1/\sigma^2 + 1/s^2} \times y$$

Thus the posterior mean is the weighted average of the prior mean and the observation, where the weights are the proportions of the precisions to the posterior precision.

Simple updating rule for Normal family

This updating rule also holds for the flat prior. The flat prior has infinite variance, so it has zero precision. The posterior precision will equal the prior precision

$$\frac{1}{\sigma^2} = 0 + \frac{1}{\sigma^2},$$

and the posterior variance equals the observation variance 2. The flat prior doesn't have a well-defined prior mean. It could be anything. We note that

$$\frac{0}{1/\sigma^2} \times \text{anything} + \frac{1/\sigma^2}{1/\sigma^2} \times y = y,$$

so the posterior mean using flat prior equals the observation y .

Predictive Density for next observation

Let y_{n+1} be the next random variable drawn after the random sample $y_1, y_2, \dots, y_n$. The predictive density of $y_{n+1} \mid y_1, y_2, \dots, y_n$ is the conditional density:

$$f(y_{n+1} \mid y_1, y_2, \dots, y_n).$$

The conditional distribution we want is found by integrating μ out of the joint posterior distribution:

$$f(y_{n+1} \mid y_1, y_2, \dots, y_n) = \int f(y_{n+1} \mid \mu) \times g(\mu \mid y_1, y_2, \dots, y_n) d\mu.$$

Predictive Density for next observation

After a "little bit" of calculus and algebra, we have that:

$$f(y_{n+1} \mid y_1, y_2, \dots, y_n) \propto \exp \left(-\frac{1}{2(\sigma^2 + s_n^2)} (y_{n+1} - m_n)^2 \right),$$

We recognize this as a Normal density with mean m_n and variance $\sigma^2 + s_n^2$, where m_n and s_n^2 denote the posterior mean and precision (after observing $y_1, y_2, \dots, y_n$). Thus, the predictive mean for the observation y_{n+1} is the posterior mean of μ given the observations $y_1, y_2, \dots, y_n$. The predictive variance is the observation variance σ^2 plus the posterior variance of μ given the observations $y_1, y_2, \dots, y_n$.

Bayesian One-sided Hypothesis Test about μ

The posterior distribution $g(\mu \mid y_1, \dots, y_n)$ summarizes our entire belief about the parameter, after viewing the data. Sometimes we want to answer a specific question about the parameter. This could be: Given the data, can we conclude the parameter μ is greater than μ_0 ? The answer to the question can be resolved by testing:

$$H_0 : \mu \leq \mu_0 \quad \text{vs} \quad H_1 : \mu > \mu_0.$$

This is an example of a one-sided hypothesis test.

Bayesian One-sided Hypothesis Test about μ

Testing a one-sided hypothesis in Bayesian statistics is done by calculating the posterior probability of the null hypothesis. When the posterior distribution $g(\mu \mid y_1, y_2, \dots, y_n)$ is $\text{Normal}(m', (s')^2)$, this can easily be found from Standard Normal tables:

$$P(H_0 : \mu \leq \mu_0 \mid y_1, \dots, y_n) = P\left(\frac{\mu - m'}{s'} \leq \frac{\mu_0 - m'}{s'}\right) = P(Z \leq \frac{\mu_0 - m'}{s'})$$

where Z is a Standard Normal random variable.

MCMC Sampling

We want to find a Markov chain that has the target distribution as its long-run distribution. Thus the support of the target will be the state space of the Markov chain. We know that the long-run distribution of a ergodic Markov chain is a solution of the steady state equation. That means that the long-run distribution $\mathbf{P}$ of a finite ergodic Markov chain with one-step transition matrix P satisfies the equation:

$$\pi = \mathbf{P}\pi.$$

MCMC Sampling

The comparable steady state equation that $\pi(\theta)$, the long-run distribution of a Markov chain with a continuous state space, satisfies is given by

$$\int_A \pi(\theta) d\theta = \int \pi(\theta) P(\theta, A) d\theta$$

for all A where $P(\theta, A)$ is the transition kernel of the chain.

Finding a MC that has the target as its long-run distribution

We need to find a probability transition kernel $P(\theta, A)$ that satisfies

$$\int g(\theta)P(\theta, A)d\theta = \int_A g(\theta)d\theta \text{ for all } A,$$

where we know that $g(\theta)$ except for the scale factor needed to make it an exact density.

Theorem

Let $q(\theta, \theta')$ be a candidate distribution that generates a candidate θ' given starting value θ . If for all θ, θ' the candidate distribution $q(\theta', \theta)$ satisfies the reversibility condition

$$g(\theta) \times q(\theta, \theta') = g(\theta') \times q(\theta', \theta)$$

then $g(\theta)$ is the long-run distribution for the Markov Chain with probability kernel

$$P(\theta, A) = \int_A q(\theta, \theta') d\theta' + r(\theta) \delta_A(\theta)$$

where $r(\theta) = 1 - \int q(\theta, \theta') d\theta'$ is the probability the chain remains at θ , and where $\delta_A(\theta)$ is the indicator function of set A .

Metropolis-Hastings Algorithm for a single parameter

Unfortunately, most candidate distributions don't satisfy the reversibility condition. For some θ and θ'

$$g(\theta)q(\theta, \theta') \neq g(\theta')q(\theta', \theta)$$

the probability of moving from θ to θ' is not the same as the probability of moving in the reverse direction. Metropolis et al. (1953) supplied the solution. They restored the balance by introducing a probability of moving

$$\alpha(\theta, \theta') = \min \left(1, \frac{g(\theta')q(\theta', \theta)}{g(\theta)q(\theta, \theta')} \right).$$

(The algorithm only requires that we know the unscaled target).

Steps of Metropolis-Hastings Algorithm

1. Start at an initial value $\theta^{[0]}$.
2. Do for $n = 1, \dots, N$.
 - a) Draw θ' from $q(\theta^{[n-1]}, \theta')$.
 - b) Calculate the probability $\alpha(\theta^{[n-1]}, \theta')$.
 - c) Draw u from $U(0, 1)$.
 - d) If $u < \alpha(\theta^{[n-1]}, \theta')$ then let $\theta^{[n]} = \theta'$, else let $\theta^{[n]} = \theta^{[n-1]}$.

Single Parameter with a Random-Walk Candidate Density

Metropolis et al. (1953) considered Markov chains with a random-walk candidate distribution. For a random-walk candidate generating distribution, the candidate is drawn from a symmetric distribution centered at the current value. Thus the candidate density is given by

$$q(\theta, \theta') = q_1(\theta' - \theta)$$

where $q_1()$ is a function symmetric about 0. Because of the symmetry $q_1(\theta' - \theta) = q_1(\theta - \theta')$, so for a random-walk candidate density, the acceptance probability simplifies to be

$$\alpha(\theta, \theta') = \min \left(1, \frac{g(\theta')q(\theta', \theta)}{g(\theta)q(\theta, \theta')} \right) = \min \left(1, \frac{g(\theta')}{g(\theta)} \right).$$

Example

Suppose we have an unscaled target density given by

$$g(\theta) = 0.8 \times e^{-\frac{1}{2}\theta^2} + 0.2 \times \frac{1}{2}e^{-\frac{(\theta-3)^2}{2 \times 2^2}}$$

Single Parameter with an Independent Candidate Density

Hastings (1970) introduced Markov chains with candidate generating density that did not depend on the current value of the chain. These are called **independent** candidate distribution

$$q(\theta, \theta') = q_2(\theta')$$

for some function $q_2(\theta)$ that must dominate the target in the tails. (We can make sure this requirement is met by graphing logarithms of the target and the candidate density).

Single Parameter with an Independent Candidate Density

For an independent candidate density, the acceptance probability simplifies to be

$$\alpha(\theta, \theta') = \min \left(1, \frac{g(\theta')q(\theta', \theta)}{g(\theta)q(\theta, \theta')} \right) = \min \left(1, \frac{g(\theta')}{g(\theta)} \times \frac{q_2(\theta)}{q_2(\theta')} \right).$$

Example

Suppose we have an unscaled target density given by

$$g(\theta) = 0.8 \times e^{-\frac{1}{2}\theta^2} + 0.2 \times \frac{1}{2}e^{-\frac{(\theta-3)^2}{2 \times 2^2}}$$

What is Prior

- Prior probability, in Bayesian Statistics, is the probability of an event before new data is collected. This is the best rational assessment of the probability of an outcome based on the current knowledge before an experiment is performed.

Type of Priors

- *Subjective Prior vs Objective Prior:*
- Subjective Prior: The prior distribution is chosen based on your beliefs about the behavior of parameter.
- Objective Prior: The prior distributions chosen in a systematic way to minimize personal bias in the result.

Type of Priors

- *Informative Prior vs Uninformative Prior:*
- Informative Prior: The prior distribution is specifically chosen based on available information or knowledge to help make the posterior distribution more precise.
- Uninformative Prior: The prior is intentionally chosen to be vague so that it plays only a minimal role in the posterior distribution. 'Let the data speak for itself.'

Type of Priors

- *Proper Prior vs Improper Prior:*
- Proper Prior: The proper distribution is a valid statistical distribution. A proper prior integrates (adds up) to 1 across all possible values of the parameter. This ensures a valid probability distribution where the sum of the probabilities for all possible outcomes is equal to 1 (certainty).
- Improper Prior: The Prior is **NOT** a valid statistical distribution (usually does not integrate to 1). It is simply a function of the parameter.

Type of Priors

- *Conjugate Prior vs Non-conjugate Prior:*
- Conjugate Prior: If the prior and the posterior belong to the same distribution family, then the prior is said to be conjugate for the likelihood.
- Non-conjugate Prior: Any prior that's not conjugate.

Type of Priors

Likelihood	Prior	Posterior
Binomial	Beta	Beta
Negative Binomial	Beta	Beta
Poisson	Gamma	Gamma
Geometric	Beta	Beta
Exponential	Gamma	Gamma
Normal (mean unknown)	Normal	Normal
Normal (variance unknown)	Inverse Gamma	Inverse Gamma
Normal (mean and variance unknown)	Normal/Gamma	Normal/Gamma
Multinomial	Dirichlet	Dirichlet

Conjugate Prior

- If the posterior distribution $P(\theta|X)$ are in the same family as the prior probability distribution $P(\theta)$, the prior and posterior are then called conjugate distributions, and the prior is called a conjugate prior for the likelihood function $P(X|\theta)$

Conjugate Prior

- Parameter: $\pi, 0 \leq \pi \leq 1$
- Likelihood: $P(y|\pi) \sim \text{Bin}(n, \pi)$
 $P(y|\pi) = \binom{n}{y} \pi^y (1 - \pi)^{(n-y)}, y = 0, 1, 2, \dots, n$
- Prior: $P(\pi) \sim \text{Beta}(\alpha, \beta)$
 $P(\pi) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{(\alpha-1)} (1 - \pi)^{(\beta-1)}$
- Posterior: $P(\pi|y)$

Conjugate Prior

$$\begin{aligned} P(\pi|y) &= \frac{P(\pi, y)}{P(y)} \\ &= \frac{P(y|\pi)P(\pi)}{P(y)} \\ &= \frac{P(y|\pi)P(\pi)}{\int_0^1 P(\pi, y) d\pi} \\ &= \frac{P(y|\pi)P(\pi)}{\int_0^1 P(y|\pi)P(\pi) d\pi} \end{aligned}$$

Conjugate Prior

$$\begin{aligned}
 \text{Denominator} &= \int_0^1 \binom{n}{y} \pi^y (1 - \pi)^{(n-y)} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{(\alpha-1)} (1 - \pi)^{(\beta-1)} d\pi \\
 &= \binom{n}{y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \pi^{(\alpha+y)-1} (1 - \pi)^{(n+\beta-y)-1} d\pi
 \end{aligned}$$

Conjugate Prior

- Let $\alpha^* = \alpha + y$ and $\beta^* = n + \beta - y$, we can get

$$\begin{aligned}
 \text{Denominator} &= \binom{n}{y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \pi^{\alpha^*-1} (1 - \pi)^{\beta^*-1} d\pi \\
 &= \binom{n}{y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha^*)\Gamma(\beta^*)}{\Gamma(\alpha^* + \beta^*)} \int_0^1 \frac{\Gamma(\alpha^* + \beta^*)}{\Gamma(\alpha^*)\Gamma(\beta^*)} \pi^{\alpha^*-1} (1 - \pi)^{\beta^*-1} d\pi \\
 &= \binom{n}{y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha + y)\Gamma(n + \beta - y)}{\Gamma(\alpha + \beta + n)}
 \end{aligned}$$

Conjugate Prior

$$\begin{aligned}
 P(\pi|y) &= \frac{\binom{n}{y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{(\alpha+y)-1} (1-\pi)^{(n+\beta-y)-1}}{\binom{n}{y} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+y)\Gamma(n+\beta-y)}{\Gamma(\alpha+\beta+n)}} \\
 &= \frac{\Gamma(\alpha+\beta+n)}{\Gamma(\alpha+y)\Gamma(n+\beta-y)} \pi^{(\alpha+y)-1} (1-\pi)^{(n+\beta-y)-1}
 \end{aligned}$$

with $\text{Beta}(\alpha^* = \alpha + y, \beta^* = n + \beta - y)$

Uninformative Prior

- If prior information about the parameter θ is available, it should be incorporated in the prior density.
- If we have no prior information, we want a prior with minimal influence on the inference. We call such a prior an uninformative prior.

Uninformative Prior(Uniform Prior)

- Definition: A uniform prior assigns equal probability to all values of the parameter within a specified interval, reflecting complete ignorance about which values are more likely.
- For a parameter θ in the interval $[a, b]$, the uniform prior is given by:

$$\pi(\theta) = \frac{1}{b - a} \quad \text{for } \theta \in [a, b]$$

Uninformative Prior(Uniform Prior)

- Suppose we are estimating the probability p of a coin landing heads. We have no prior knowledge about the coin, so we start with a uniform prior.
- Prior: Since we are completely ignorant about the coin's bias, we use a uniform prior over the interval $[0, 1]$

$$P(p) \sim \text{Uniform}(0, 1)$$

Uninformative Prior(Uniform Prior)

- Suppose we flip the coin 10 times and observe 7 heads and 3 tails. The likelihood of observing this data given the parameter p is given by the binomial distribution

$$\binom{10}{7} p^7 (1-p)^3$$

- Our posterior would be

$$P(p|data) = 1 \times \binom{10}{7} p^7 (1-p)^3$$

Uninformative Prior(Uniform Prior)

- The problem of re-parametrization: let $\theta = p^2$?
- Jacobian Determinant: $J = \left| \frac{dp}{d\theta} \right| = \left| \frac{d\sqrt{\theta}}{d\theta} \right| = \frac{1}{2\sqrt{\theta}}$
- Transformed Prior = $P(p) \times J = \frac{1}{2\sqrt{\theta}}$
- $P(\theta) \propto \frac{1}{2\sqrt{\theta}}$, the uniform prior on P does not remain uniform after reparameterization.

Uninformative Prior(Jeffreys Prior)

- Definition: The Jeffreys prior is a type of uninformative prior that is invariant under reparameterization. It is based on the Fisher information, ensuring that the prior remains the same regardless of how the parameter is transformed.
- For a parameter θ , the Jeffreys prior is given by:

$$\pi(\theta) \propto \sqrt{I(\theta)}$$

- Where $I(\theta)$ is the Fisher information:

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2 \log L(\theta)}{\partial \theta^2} \right]$$

Lack of Objectivity

In our analyses, we desire objectivity so that we can form some sort of generalisation as well as to minimise bias. Flat priors do not necessarily ensure objectivity within our study. There are many reasons for this:

- Dependence on parameterization
- Conceptual Issues
- Arbitrariness

If flat priors are not objective, then what could be?

Relevance To Our Study

Why exactly are we interested in Jeffreys' Prior? **Recall:**

- Our main goal is to refine Bayesian analyses with an **Improper** prior $\pi(\theta)$ and a set of data $y_1, \dots, y_n$.
- As noted previously, the specific process involves splitting the dataset into l and m parts: $y_1, \dots, y_l$ and $y_1^*, \dots, y_m^*$, where $n = l + m$
- To achieve our goal, we need a general way to achieve an objective uninformative prior which Jeffreys prior will allow us to do.

Binomial Model Re-visit

Recall our Binomial sampling model:

$$\pi(y|\theta) = \text{Bin}(y|n, \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

How do we determine a good uninformative prior for this model? Which parameter?

- How about θ ? A flat $Unif(0,1)$?
- What about $\phi = \frac{\theta}{1-\theta}$ (i.e the odds)? $Unif(0, \infty)$?
- The $\rho = \log(\frac{\theta}{1-\theta})$ (i.e the log-odds)? $Unif(-\infty, \infty)$?

Using a Uniform prior here depends on how we parameterise.

Jeffreys' Prior

Harold Jeffreys suggested an interesting solution to ensure objectivity.

Definition

Given Fisher Information $J(\theta)$ for a model with one parameter, Jeffreys prior is the prior that satisfies $\pi(\theta) \propto J(\theta)^{1/2}$

- Where, $J(\theta) = -\mathbb{E} \left(\left(\frac{d \log(\pi(y|\theta))}{d\theta} \right)^2 \middle| \theta \right) = -\mathbb{E} \left(\frac{\partial^2 \log \pi(y|\theta)}{\partial \theta^2} \right)$
- **Note:** The Fisher information quantifies how much info the observed data provides about θ in a statistical model.
- Intuitively, Jeffreys prior assigns more prior weight to parameter values where the Fisher Information is higher.

Jeffreys' Prior Cont...

- This indicates that those values are more informative or carry more information about the parameters.
- Objective in the sense that any change-of-parameter yields the Jeffreys prior for that parameter.
- When the parameter space is unbounded, Jeffreys' prior is typically improper.

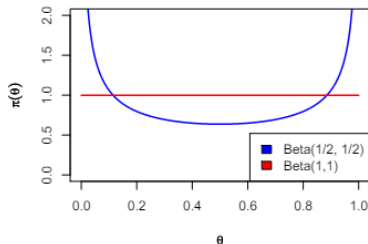


Figure 1: Jeffreys prior and flat prior densities

Single-parameter Invariance

- Jeffreys' prior is **invariant** since when we express the model in terms of a different parameterization, the Jeffreys' prior remains unchanged.

- Mathematically, we say prior $\pi(\theta)$ is invariant when $\pi(\phi) = \pi(\theta) \left| \frac{d\phi}{d\theta} \right|$

- Given that the Fisher data changes during reparameterization as,

$$J(\phi) = J(\theta) \left| \frac{d\phi}{d\theta} \right|$$

- We then define $\pi(\phi) \propto \sqrt{J(\phi)}$ and similarly with $\pi(\theta)$ which thus yields our desired invariance.

Examples

- Assuming data $\sim \text{Bin}(n, \theta) \implies J(\theta) = \frac{n}{\theta(1-\theta)}$. Since n is constant, Jeffreys prior is $\pi(\theta) \propto \sqrt{\frac{1}{\theta(1-\theta)}}$
 - For a conjugate Beta family: $\text{Beta}(\frac{1}{2}, \frac{1}{2})$
- If data $\sim \text{Poisson}(\mu) \implies J(\mu) = \frac{1}{\mu}$ and Jeffreys prior is $\pi(\mu) \propto \frac{1}{\sqrt{\mu}}$
 - For a conjugate Gamma family: $\text{Gamma}(\frac{1}{2}, 0)$
- Suppose data $\sim N(\mu, \sigma^2)$, where σ^2 is known $\implies J(\mu) = 1$ and Jeffreys prior is $\pi(\theta) \propto 1$
- If data $\sim N(\mu, \sigma^2)$, where μ is known $\implies J(\sigma^2) = \frac{1}{2(\sigma^2)^2}$ and Jeffreys prior is $\pi(\theta) \propto \frac{1}{\sigma^2}$

Poisson Jeffreys Prior

Theorem

Given data $\sim \text{Poisson}(\mu)$, the Jeffreys prior for Poisson likelihood follows

$$\pi(\theta) \propto \frac{1}{\sqrt{\mu}}$$

Proof:

Poisson Jeffreys Prior

Proof cont.

Binomial Jeffreys Prior

Theorem

Given data $\sim \text{Bin}(n, \theta)$, the Jeffreys prior for Binomial likelihood follows
$$\pi(\theta) \propto \theta^{-1/2}(1 - \theta)^{-1/2}$$

Proof:

Binomial Jeffreys Prior

Proof cont.

Normal Jeffreys Prior (σ^2 is known)

Theorem

Given data $\sim N(\mu, \sigma^2)$, where σ^2 is known, the Jeffreys prior is $\pi(\mu) \propto 1$

Proof:

$$\log L(\mu|y) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y-\mu)^2}{2\sigma^2}$$

$$\frac{\partial}{\partial \mu} \log L(\mu|y) = \frac{y-\mu}{\sigma^2}$$

$$\frac{\partial^2}{\partial \mu^2} \log L(\mu|y) = -\frac{1}{\sigma^2}$$

$$J(\mu) = -\mathbb{E} \left[\frac{\partial^2}{\partial \mu^2} \log L(\mu|y) \right] = \frac{1}{\sigma^2}$$

$$\pi(\mu) \propto \sqrt{J(\mu)} = \sqrt{\frac{1}{\sigma^2}} = \frac{1}{\sqrt{\sigma^2}}$$

Since σ^2 is a known constant, $\frac{1}{\sqrt{\sigma^2}}$ is just a constant. Thus: $\pi(\mu) \propto 1$

Normal Jeffreys Prior (μ is known)

Theorem

Given data $\sim N(\mu, \sigma^2)$, where μ is known, the Jeffreys prior is $\pi(\sigma^2) \propto \frac{1}{\sigma^2}$

Proof:

$$\log L(\sigma^2|y) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y-\mu)^2}{2\sigma^2}$$

$$\frac{\partial}{\partial \sigma^2} \log L(\sigma^2|y) = -\frac{1}{2\sigma^2} + \frac{(y-\mu)^2}{2(\sigma^2)^2}$$

$$\frac{\partial^2}{(\partial \sigma^2)^2} \log L(\sigma^2|y) = \frac{1}{2(\sigma^2)^2} - \frac{(y-\mu)^2}{(\sigma^2)^3}$$

$$J(\sigma^2) = -\mathbb{E} \left[\frac{\partial^2 \log L(\sigma^2|y)}{(\partial \sigma^2)^2} \right] = \frac{1}{2(\sigma^2)^2}$$

$$\pi(\sigma^2) \propto \sqrt{J(\sigma^2)} = \sqrt{\frac{1}{2(\sigma^2)^2}} = \frac{1}{\sqrt{2}\sigma^2}$$

Thus, $\pi(\sigma^2) \propto \frac{1}{\sigma^2}$

Exercise

Suppose $\pi(y|\theta) \sim \text{Bin}(n, \theta)$. Assume we are using a Jeffreys prior for θ , that is $\pi(\theta) = \theta^{-1/2}(1 - \theta)^{-1/2}$.

- a.) Find the posterior for θ .
- b.) Find the posterior mean of θ
- c.) Find the variance of the posterior distribution of θ

Exercise cont.

Solution:

a.)

$$\begin{aligned}
 \pi(\theta|y) &\propto \pi(\theta)\pi(y|\theta) \\
 &\propto \left[\binom{n}{y} \theta^y (1-\theta)^{n-y} \right] \left[\theta^{-1/2} (1-\theta)^{-1/2} \right] \\
 &\propto \theta^{y-1/2} (1-\theta)^{n-y-1/2} \\
 &\sim \text{Beta}(\alpha^* = y + \tfrac{1}{2}, \beta^* = n - y + \tfrac{1}{2})
 \end{aligned}$$

b.)

$$\mathbb{E}[\theta|y] = \frac{\alpha^*}{\alpha^* + \beta^*} = \frac{y + \frac{1}{2}}{(y + \frac{1}{2}) + (n - y + \frac{1}{2})} = \frac{y + \frac{1}{2}}{n + 1}$$

c.)

$$V[\theta|y] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{(y + \frac{1}{2})(n - y + \frac{1}{2})}{(n+1)^2(n+2)}$$

Limitations

Although Jeffreys priors have advantages such as invariance, there are also some limitations that come with Jeffreys prior:

- It's usually Improper (which means that it does not integrate to 1). This also means that we might not be able to obtain a proper posterior in some cases.
- Some claim that Jeffreys Prior violates the likelihood principle.
- The Jeffreys prior depends on the Fisher information, which in turn depends on the specified model. In practice, any model misspecification will impact the the Jeffreys prior, which might lead to misleading inferences

Empirical Usage of Jeffreys Prior

When is it a good idea to use Jeffreys prior in practice?

- When you have little prior information about θ .
- One of your parameters for your model are unknown, you may choose to use Jefferys prior to illicit the unknown parameter, Ex: $N(\mu, \sigma^2)$.
- Utilising a Jefferys prior for a part of the dataset to develop a posterior (i.e what we will be doing for our study).
- If your analysis might involve changing the parameterisation of your model.

The Reference Prior

- Reference priors are another way to form an uninformative prior.
- It involves maximising the KL (Kullback–Leibler) divergence between the prior and the posterior.
- Maximising the KL divergence between the prior and the posterior is the same thing as maximising the mutual information between the prior and our data.
- If we choose a prior that maximises the mutual information, we are not allowing our prior to exert much of an influence on our posterior.

Jeffreys' prior and Multiparameter

- In the multiparameter case, we consider a normal distribution with unknown mean μ and variance σ^2 . Here, both mean μ and variance σ^2 are the parameters we need to estimate.

Consider a normal distribution $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$p(x|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- X is the observed data.
- μ is the mean of the distribution.
- σ^2 is the variance of the distribution.
- $\theta = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}$ is the parameter vector.

Jeffreys' prior and Multiparameter

- Jeffreys' prior is given by:

$$\pi_J(\theta) \propto |I(\theta)|^{1/2}$$

where

$$I(\theta)_{ij} = -\mathbb{E} \left[\frac{\partial^2 \log p(X|\theta)}{\partial \theta_i \partial \theta_j} \right] = -\mathbb{E} \begin{bmatrix} \frac{\partial^2 \log p(X|\theta)}{\partial \mu^2} & \frac{\partial^2 \log p(X|\theta)}{\partial \mu \partial \sigma^2} \\ \frac{\partial^2 \log p(X|\theta)}{\partial \sigma^2 \partial \mu} & \frac{\partial^2 \log p(X|\theta)}{\partial \sigma^2} \end{bmatrix}$$

θ is a parameter vector, $I(\theta)$ is the Fisher information matrix, and $|I(\theta)|$ is its determinant.

Jeffreys prior: Multiparameter case - Normal density

Theorem

Given data $\sim N(\mu, \sigma^2)$, where both μ and σ^2 are unknown, the Jeffreys prior is $\pi(\mu, \sigma^2) \propto \frac{1}{\sigma^3}$

Proof:

First, we compute the Fisher information matrix.

The log-likelihood function for a normal distribution is given by:

$$\log p(X \mid \mu, \sigma^2) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(X - \mu)^2}{2\sigma^2}$$

Jeffreys prior: Multiparameter case - Normal density

The 1_{st} derivative with respect to μ :

$$\frac{\partial \log p(X | \mu, \sigma^2)}{\partial \mu} = \frac{X - \mu}{\sigma^2}$$

The 1_{st} derivative with respect to σ^2 :

$$\frac{\partial \log p(X | \mu, \sigma^2)}{\partial \sigma^2} = -\frac{1}{2\sigma^2} + \frac{(X - \mu)^2}{2\sigma^4}$$

Jeffreys prior: Multiparameter case - Normal density

The 2_{nd} derivative:

- With respect to μ :

$$\frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial \mu^2} = -\frac{1}{\sigma^2}$$

- With respect to σ^2 :

$$\frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial (\sigma^2)^2} = \frac{1}{2\sigma^4} - \frac{(X - \mu)^2}{\sigma^6}$$

- With cross terms(μ and σ^2):

$$\frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial \mu \partial \sigma^2} = \frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial \sigma^2 \partial \mu} = 0$$

Jeffreys prior: Multiparameter case - Normal density

Taking the expected values to find the Fisher information matrix:

$$\mathbb{E} \left(\frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial \mu^2} \right) = -\frac{1}{\sigma^2}$$

$$\mathbb{E} \left(\frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial \mu \partial \sigma^2} \right) = 0$$

$$\mathbb{E} \left(\frac{\partial^2 \log p(X | \mu, \sigma^2)}{\partial (\sigma^2)^2} \right) = -\frac{1}{2\sigma^4}$$

Jeffreys prior: Multiparameter case - Normal density

Thus, the Fisher information matrix $I(\theta)$ is:

$$I(\theta) = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{pmatrix}$$

Jeffreys prior is given by:

$$\pi_J(\theta) \propto |I(\theta)|^{1/2}$$

To find the Jeffreys prior, we need the determinant of the Fisher information matrix:

$$|I(\theta)| = \left| \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{pmatrix} \right| = \frac{1}{\sigma^2} \cdot \frac{1}{2\sigma^4} = \frac{1}{2\sigma^6}$$

Thus,

$$\pi_J(\mu, \sigma^2) \propto \left(\frac{1}{2\sigma^6} \right)^{1/2} = \frac{1}{\sqrt{2}\sigma^3}$$

$$\pi_J(\mu, \sigma^2) \propto \frac{1}{\sigma^3}$$

Multiparameter problems

Problems with the Naive Jeffreys Prior:

$$\pi_J(\mu, \sigma^2) \propto \frac{1}{\sigma^3}$$

- $\frac{1}{\sigma^3}$ has poor convergence properties.
- Jeffreys himself proposed using the prior $\pi_J(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$, which is a product of the separate priors for μ and σ^2 .

$$p_J(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$$

By using this modified prior, we achieve better convergence properties and a more practical solution for parameter estimation.

Jeffreys' prior and Multivariate normal density

- Let $X \sim \mathcal{N}(\mu, \Sigma)$. Here, X is a p -dimensional vector with random variables, μ is a p -dimensional mean vector, and Σ is a $p \times p$ matrix.
- The pdf of multivariate normal distribution:

$$p(x|\theta) = \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x - \mu)^T |\Sigma|^{-1} (x - \mu)\right)$$

Jeffreys' prior and Multivariate normal density

Theorem

Given data $\sim N(\mu, \sigma)$, where both μ and σ are unknown, the Jeffreys prior is $\pi_J(\mu, \Sigma) \propto |\Sigma|^{-(p+1)/2}$.

Proof: Fisher information matrix:

For μ :

$$I_{\mu\mu}(\theta) = -\mathbb{E} \left(\frac{\partial^2 \log p(X | \mu, \Sigma)}{\partial \mu \partial \mu^T} \right) = \Sigma^{-1}$$

For Σ :

$$I_{\Sigma\Sigma}(\theta) = -\mathbb{E} \left(\frac{\partial^2 \log p(X | \mu, \Sigma)}{\partial \Sigma \partial \Sigma^T} \right) = \frac{1}{2}(\Sigma^{-1} \otimes \Sigma^{-1})$$

Where $\otimes$ denotes the Kronecker product.

For Cross Terms (μ and Σ):

$$I_{\mu\Sigma}(\theta) = I_{\Sigma\mu}(\theta) = 0$$

Combined Fisher Information Matrix:

Jeffreys' prior and Multivariate normal density

To find the Jeffreys prior, we need the determinant of the Fisher information matrix $I(\theta)$:

$$|I(\theta)| = \left| \begin{pmatrix} \Sigma^{-1} & 0 \\ 0 & \frac{1}{2}(\Sigma^{-1} \otimes \Sigma^{-1}) \end{pmatrix} \right| = |\Sigma^{-1}| \cdot \left| \frac{1}{2}(\Sigma^{-1} \otimes \Sigma^{-1}) \right|$$

Simplifying further:

$$|\Sigma^{-1}| = (\det(\Sigma))^{-1}$$

$$|\Sigma^{-1} \otimes \Sigma^{-1}| = (\det(\Sigma^{-1}))^p = (\det(\Sigma))^{-p}$$

Thus:

$$|I(\theta)| = (\det(\Sigma))^{-1} \cdot \frac{1}{2^p} (\det(\Sigma))^{-p} = \frac{1}{2^p} (\det(\Sigma))^{-(p+1)}$$

Jeffreys' prior and Multivariate normal density

Jeffreys prior is given by:

$$\pi_J(\theta) \propto |I(\theta)|^{1/2}$$

Substituting the determinant:

$$\pi_J(\mu, \Sigma) \propto \left(\frac{1}{2^p} (\det(\Sigma))^{-(p+1)} \right)^{1/2} = \frac{1}{2^{p/2}} (\det(\Sigma))^{-(p+1)/2}$$

Since $2^{p/2}$ is a constant, it can be ignored in the proportionality:

$$\pi_J(\mu, \Sigma) \propto |\Sigma|^{-(p+1)/2}$$

Multivariate problem

- The exponent $-\frac{(p+1)}{2}$ indicates how the prior scales with the dimensionality p of the multivariate normal distribution.
- In high-dimensional spaces (large p), the determinant $|\Sigma|$ is raised to a larger negative power, which can make the prior extremely sensitive to changes in Σ . This can lead to overly diffuse or flat priors in high dimensions, which might not provide sufficient regularization.

Multivariate problem

What if use the separate product of Jeffreys Prior for μ and for Σ :

$$\pi_J(\mu \mid \Sigma) \propto |\Sigma|^{-1/2}$$

$$\pi_J(\Sigma) \propto |\Sigma|^{-(p+1)/2}$$

$$\pi_J(\mu, \Sigma) = \pi_J(\mu \mid \Sigma) \pi_J(\Sigma)$$

The combined Jeffreys prior is:

$$\pi_J(\mu, \Sigma) \propto |\Sigma|^{-1/2} \cdot |\Sigma|^{-(p+1)/2} = |\Sigma|^{-(p+2)/2}$$

However, the result still has the same problem

$$\pi_J(\mu, \Sigma) \propto |\Sigma|^{-(p+2)/2}$$

Reference Prior and Jeffreys prior

Definition of jeffreys prior and reference prior

- **Jeffreys prior**

Jeffreys priors are based on the Fisher information matrix, which measures the amount of information that an observable random variable carries about an unknown parameter.

- **Reference prior**

Reference priors are designed to maximize the information gain from the data. They are derived to maximize the divergence (e.g., Kullback-Leibler divergence) between the posterior and prior distributions, allowing the data to have the maximum impact on the posterior.

Reference Prior

Advantages of reference prior in highdimensional case:

- **1. Easy computation**

Calculating Jeffreys priors in high-dimensional settings is complex. The Fisher Information Matrix (FIM) needs to be computed and its determinant evaluated, which becomes increasingly difficult as the dimensionality increases.

Reference priors are designed to maximize the information gain and can be more straightforward to calculate in high-dimensional spaces. They often involve optimizing mutual information.

Reference Prior

- **2. Proper priors** Jeffreys priors can be improper in high dimensions, meaning they do not integrate to a finite value. Reference priors are typically constructed to ensure propriety, meaning they integrate to 1 and lead to proper posteriors.
- **3. Regularization** Jeffreys Priors is often too flat, providing insufficient regularization. This can leave the posterior overly diffuse and under-constrained, making reliable inference difficult. Reference Priors is designed to balance non-informativeness with enough regularization to ensure well-behaved posteriors. This helps in constraining the parameter estimates in high-dimensional spaces.

Reference Prior

Given data X parameterized by Θ with a sufficient statistic $T = T(X)$, the reference prior $p(\theta)$ maximizes the mutual information between θ and T . The K-L divergence between the posterior $p(\theta | t)$ and the prior $p(\theta)$, averaged over the distribution of T , is given by:

$$\int p(\theta | t) \log \frac{p(\theta | t)}{p(\theta)} d\theta$$

Its expectation over the distribution of T is:

$$I(\Theta, T) = \int p(t) \left(\int p(\theta | t) \log \frac{p(\theta | t)}{p(\theta)} d\theta \right) dt$$

Which can be rewritten as:

$$I(\Theta, T) = \int \int p(\theta, t) \log \frac{p(\theta | t)}{p(\theta)p(t)} d\theta dt$$

Reference Prior

This expression is recognized as the mutual information between θ and t . Therefore, choosing a reference prior involves finding $p^*(\theta)$ that maximizes the mutual information:

$$p^*(\theta) = \arg \max_{p(\theta)} I(\Theta, T)$$

Overview: Why do we Cross-Validate

- What happens if the models are trained with no Cross-Validation?
- Assessing Model Accuracy: Estimate how accurately a predictive model will perform.
- No one method dominates all others over all possible data sets. On a particular data set, one specific method may work best, but some other methods may work better on a similar but different data set.
- We use *Mean Square Error* as a metric to measure the quality of the fit.

$$\mathbf{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

Traditional Cross-Validation Technique

The Validation Set Approach: A set of n observations is randomly split into a training set and a validation set. The statistical learning method is fit on the training set, and its performance is evaluated on the validation set. Note that this partition can be 50-50 or 75-25, the rule of thumb is leaning towards 75% training data and 25% testing data.

What's the drawback?

k-fold Cross-Validation

- Shuffle the dataset randomly.
- Split the dataset into k groups.
- Take the group as a hold-out or test data set.
- Take the remaining groups as a training data set.
- Fit a model on the training set and evaluate it on the test set.
- Summarize the model's skill using the sample of model evaluation scores.

Importantly, each observation in the data sample is assigned to an individual group and stays in that group for the duration of the procedure. This means that each sample can be used in the hold-out set 1 time and to train the model $k-1$ times.

$$CV_k = \frac{1}{k} \sum_{i=1}^k MSE_i$$

Cross-Validation

- First split the data into two parts by leaving out the **first** observation:
 - a validation set: (x_1, y_1)
 - a training set: the remaining observations $(x_2, y_2), \dots, (x_n, y_n)$
 - using the training set, we fit the model $\hat{f}_1$ and predict y_1 as $\hat{f}_1(x_1)$ using the value x_1 . The test error could be approximated by

$$\text{MSE}_1 = (y_1 - \hat{f}_1(x_1))^2.$$

- not good enough!

Cross-Validation

- Repeat the procedure by leaving out the **second** observation:
 - A validation set: (x_2, y_2)
 - A training set: the remaining observations $(x_1, y_1), (x_3, y_3), \dots, (x_n, y_n)$
 - Using the training set, we fit the model $\hat{f}_2$ and predict y_2 as $\hat{f}_2(x_2)$ using the value x_2 .
 - Compute:

$$\text{MSE}_2 = (y_2 - \hat{f}_2(x_2))^2.$$

- Repeating the approach n times by leaving out each observation to obtain $\text{MSE}_1, \dots, \text{MSE}_n$.
- The LOOCV estimate for the test MSE is the average of these n test error estimates:

$$\text{CV}(n) = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i.$$

Cross-Validation

- **k-fold CV** is to randomly divide the data into k (roughly) equal-sized groups or folds.
- The first fold is treated as a validation set, and the method is fit on the remaining $k - 1$ folds. We compute the mean squared error, MSE_1 , for the observations in the first fold.
- Then we repeat the procedure to fold 2, fold 3, $\dots$, fold k , and get $\text{MSE}_2, \text{MSE}_3, \dots, \text{MSE}_k$.
- The k-fold CV estimate is computed by averaging these values:

$$\text{CV}(k) = \frac{1}{k} \sum_{i=1}^k \text{MSE}_i.$$

Remark

- LOOCV is a special case of n -fold CV.
- 5-fold or 10-fold is commonly used in practice.

Cross-Validation

- Cross-validation also works for classification problems.
- For LOOCV, we split the data in the same way as before. We compute the error on the validation set:

$$\text{Err}_1 = 1\{y_1 \neq \hat{f}_1(x_1)\}.$$

- Then we repeat the procedure n times, and get $\text{Err}_2, \text{Err}_3, \dots, \text{Err}_n$.
- The LOOCV estimate is computed by averaging these values:

$$\text{CV}(n) = \frac{1}{n} \sum_{i=1}^n \text{Err}_i.$$

Derive Normal(θ , 1) Jeffreys Prior

- For a parameter θ , the Jeffreys prior is given by:

$$\pi(\theta) \propto \sqrt{I(\theta)}$$

- and Fisher information:

$$I(\theta) = -\mathbb{E} \left[\frac{\partial^2 \log L(\theta)}{\partial \theta^2} \right]$$

- Likelihood: $\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \theta)^2}{2\sigma^2}\right]$
 $= (2\pi)^{-\frac{n}{2}} \times \exp\left[-\frac{1}{2} \sum_{i=1}^n (y_i - \theta)^2\right]$

Derive Normal(θ , 1) Jeffreys Prior

- $\ln f(y_i|\theta)$: $-\frac{n}{2}\ln(2\pi) - \frac{1}{2}\sum_{i=1}^n (y_i - \theta)^2$
 $= -\frac{n}{2}\ln(2\pi) - \frac{1}{2}\sum_{i=1}^n y_i^2 + \theta\sum_{i=1}^n y_i - \frac{1}{2}\theta^2 n$
- $\frac{\partial^2}{\partial \theta^2} \ln f(y_i|\theta) = -n$
- Thus, $I(\theta) = -\mathbb{E}(-n) = n$
- And we can get our Jeffreys Prior $\sqrt{I(\theta)} = \sqrt{n} \propto 1$

Using conjugate prior

- According to what have learned so far, we know that if we have a Beta prior and Normal likelihood with known variance, we will get a Normal distribution that follows $N(\theta, \frac{1}{n})$
- Since our prior is 1, we can use $\text{Beta}(1, 1)$ as our prior.
- To check if this is a proper prior, we need to integrate it over the entire parameter space. The integral is given by:

$$\int_{-\infty}^{\infty} p(\theta) d\theta = \int_{-\infty}^{\infty} 1 d\theta$$

Evaluating this integral, we get:

$$\int_{-\infty}^{\infty} 1 d\theta = \infty$$

Since the integral does not converge to 1 but instead diverges to infinity, this indicates that the prior $p(\theta) = 1$ is an improper prior.

KL divergence between two normal

Let $P \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Q \sim \mathcal{N}(\mu_2, \sigma_2^2)$. The probability density functions are:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(x - \mu_1)^2}{2\sigma_1^2}\right)$$

$$q(x) = \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(x - \mu_2)^2}{2\sigma_2^2}\right)$$

The KL divergence $D_{KL}(P \parallel Q)$ is given by:

$$D_{KL}(P \parallel Q) = \int_{-\infty}^{\infty} p(x) \log \frac{p(x)}{q(x)} dx$$

After the calculation, we can get the final expression:

$$D_{KL}(P \parallel Q) = \log\left(\frac{\sigma_2}{\sigma_1}\right) + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

Bayesian information criterion (BIC)

- In statistics, the Bayesian information criterion (BIC) is a criterion for model selection among a finite set of models; models with lower BIC are generally preferred.
- The formula is $BIC = -2 \ln(L) + k \ln(n)$ where:
 - L is the likelihood of the model given the data.
 - k is the number of parameters in the model.
 - n is the number of data points.

Poisson-Gamma Example

- Recall that, given $y_1, y_2, \dots, y_n$ is an i.i.d sample from $Poisson(\theta)$, the jeffreys prior is $\pi(\theta) \propto \frac{1}{\sqrt{\mu}}$
- How do we actually use this non-informative prior?
- What does the K-Fold cross validation process look like with poisson data?

Poisson-Gamma Example

- Before we begin, let's derive the Posterior that we would get if we used the **entire** poisson data to get a posterior:

Bayesian updating process - Poisson data

- **First:** Partition the data into K equally folds. The notation we use for this is $D_1, D_2, \dots, D_K$.
- **Second:** Derive the posterior distribution for the j^{th} fold:

Bayesian updating process cont. - Poisson data

- **Third:** Use the Interim posterior as the new prior for the final posterior distribution for the j^{th} fold:

KL Divergence - Poisson data

- We desire for the smallest KL Divergence to ensure we are closest to our true posterior.
- In R, you can compute this as:

```
# Function to compute KL divergence between two probability distributions
compute_kl_divergence <- function(p, q) {
  k1 <- sum(p * log(p / q))
  return(k1)
}
```

- 'p' represents the probabilities from the true posterior distribution, which is calculated using a large poisson simulated dataset.
- 'q' represents the probabilities from the posterior distribution obtained from each fold of the cross-validation.
- In the Poisson case, you will notice that the KL divergence you compute will be 0 (or very close to it). Why is this the case?

Summary: Refine Bayesian analyses using Cross validation

Main Steps:

- **1. Split Data:** Using K-fold cross validation to partition data $y_1, \dots, y_n$ into K folds, $D_1, D_2, \dots, D_k$
- **2. Stage1:**

$$\pi^{(k)}(\theta \mid D_I^{(k)}) \propto \pi(\theta) \prod_{y \in D_I^{(k)}} p(y \mid \theta).$$

, where $D_I^{(k)} = D_k$, Jeffreys prior $\pi(\theta)$, sample likelihood $p(y \mid \theta)$, interim posterior $\pi^{(k)}(\theta \mid D_I^{(k)})$

- **3. Stage2:**

$$\pi^{\star(k)}(\theta \mid D_{\parallel}^{(k)}) \propto \pi^{(k)}(\theta \mid D_I^{(k)}) \prod_{y \in D_{\parallel}^{(k)}} p(y \mid \theta).$$

, where $D_{\parallel}^{(k)} = \bigcup_{i \neq k} D_i$, prior(interim posterior) $\pi^{(k)}(\theta \mid D_I^{(k)})$, sample likelihood $p(y \mid \theta)$, final posterior $\pi^{\star(k)}(\theta \mid D_{\parallel}^{(k)})$

Summary: Refine Bayesian analyses using Cross validation

- **4. Repeat stage1 and stage2 K times**, get K modified priors and posteriors.
- **5. Evaluate Performance**
 - **KL Divergence:** Compute the KL divergence between the posterior distribution based on the Jeffreys prior using the full dataset and K posteriors based on the K folds. The smaller the KL divergence, the closer the two distributions are to each other. If KL divergence equal to 0, it means two distributions are exactly the same.
 - **MSE:** Fixed one fold of data as a validation set. Compute the mean square error between the validation data and the mean of posterior predictive.

$$\text{MSE} = \frac{1}{n_{\text{val}}} \sum_{i=1}^{n_{\text{val}}} (X_{\text{val},i} - \mu_{\text{predpost}})^2$$

The smaller the MSE, the better the predictive accuracy of the posterior.

Summary: Different Distributions

Let $y_1, \dots, y_n \sim \text{Poisson}(\theta)$:

- **Jeffreys prior:** $\pi(\theta) \propto \frac{1}{\sqrt{\theta}} \sim \text{Gamma}(\alpha = 0.5, \beta = 0)$
- **Interim posterior:** $\pi^{(k)}(\theta \mid D_I^{(k)}) \sim \text{Gamma}(\alpha' = 0.5 + \sum_{i=1}^{n_1} x_i, \beta' = n_1)$
- **Final posterior:** $\pi^{*(k)}(\theta \mid D_{\parallel}^{(k)}) \sim \text{Gamma}(\alpha'' = \alpha' + \sum_{i=1}^{n_2} x_i, \beta'' = \beta' + n_2)$
- **KL divergence** $D_{KL}(\Gamma(\alpha_1, \beta_1) \parallel \Gamma(\alpha_2, \beta_2))$:

$$(\alpha_1 - \alpha_2)\psi(\alpha_1) - \log \frac{\Gamma(\alpha_1)}{\Gamma(\alpha_2)} + \alpha_2 \log \frac{\beta_1}{\beta_2} + \alpha_1 \frac{\beta_2 - \beta_1}{\beta_1}$$

where ψ is the digamma function, which is the derivative of the logarithm of the gamma function, and Γ is the gamma function.

- **mean of predictive posterior:** $\mu = \frac{\alpha'' + \sum x_i}{\beta'' + n_2}$

Summary: Different Distributions

Let $y_1, \dots, y_n \sim N(\theta, \sigma^2 = 1)$, θ is unknown:

- **Jeffreys prior:** $\pi(\theta) \propto 1$
- **Interim posterior:** $\pi^{(k)}(\theta \mid D_I^{(k)}) \sim N(\mu' = \bar{y}_1, \sigma^{2'} = \frac{1}{n_1})$
- **Final posterior:** $\pi^{*(k)}(\theta \mid D_{II}^{(k)}) \sim N(\mu'' = \frac{\bar{y}_2 n_2 + \frac{\mu'}{\sigma^{2'}}}{n_2 + \frac{1}{\sigma^{2'}}}, \sigma^{2''} = \frac{1}{n_2 + \frac{1}{\sigma^{2'}}})$,
where n_1 refers to the number of dataset in stage1, $\bar{y}_1$ is the mean of dataset in stage1; n_2 refers to the number of dataset in stage2, $\bar{y}_2$ is the mean of dataset in stage2.
- **KL divergence** $KL(N(\mu_1, \sigma_1^2) \parallel N(\mu_2, \sigma_2^2))$:

$$\log \left(\frac{\sigma_2}{\sigma_1} \right) + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

- **Predictive posterior:**

$$y_{n+1} \mid y \sim \mathcal{N}(\mu'', \sigma^{2''})$$

Summary: Different Distributions

Let $y_1, \dots, y_n \sim \text{Exp}(\theta)$:

- **Jeffreys prior:** $\pi(\theta) \propto \frac{1}{\theta} \sim \text{Gamma}(\alpha = 0, \beta = 0)$
- **Interim posterior:** $\pi^{(k)}(\theta \mid D_I^{(k)}) \sim \text{Gamma}(\alpha' = n_1, \beta' = \sum_{i=1}^{n_1} x_i)$
- **Final posterior:** $\pi^{\star(k)}(\theta \mid D_{\parallel}^{(k)}) \sim \text{Gamma}(\alpha'' = \alpha' + n_2, \beta'' = \beta' + \sum_{i=1}^{n_2} x_i)$
- **KL divergence** $D_{KL}(\Gamma(\alpha_1, \beta_1) \parallel \Gamma(\alpha_2, \beta_2))$:

$$(\alpha_1 - \alpha_2)\psi(\alpha_1) - \log \frac{\Gamma(\alpha_1)}{\Gamma(\alpha_2)} + \alpha_2 \log \frac{\beta_1}{\beta_2} + \alpha_1 \frac{\beta_2 - \beta_1}{\beta_1}$$

- **mean of predictive posterior:** $\mu = \frac{\alpha'' + n_2}{\beta'' + \sum_{i=1}^n x_i}$

Summary: Different Distributions

Let $y_1, \dots, y_n \sim \text{Bin}(n, \theta)$:

- **Jeffreys prior:** $\pi(\theta) \propto \frac{1}{\sqrt{\theta(1-\theta)}} \sim \text{Beta}(\alpha = \frac{1}{2}, \beta = \frac{1}{2})$
- **Interim posterior:** $\pi^{(k)}(\theta \mid D_I^{(k)}) \sim \text{Beta}(\alpha' = \frac{1}{2} + y, \beta' = n - y + \frac{1}{2})$
- **Final posterior:** $\pi^{*(k)}(\theta \mid D_{II}^{(k)}) \sim \text{Gamma}(\alpha'' = \alpha' + y, \beta'' = \beta' + n - y)$
- **KL divergence** $D_{KL}(\mathbf{Beta}(\alpha_1, \beta_1) \parallel \mathbf{Beta}(\alpha_2, \beta_2))$

$$\log \frac{B(\alpha_2, \beta_2)}{B(\alpha_1, \beta_1)} + (\alpha_1 - \alpha_2)\psi(\alpha_1) - (\alpha_1 - \alpha_2)\psi(\alpha_1 + \beta_1) + (\beta_1 - \beta_2)\psi(\beta_1)$$

- $(\beta_1 - \beta_2)\psi(\alpha_1 + \beta_1)$, where ψ is the digamma function.

- **mean of predictive posterior:** $\mu = \frac{\alpha''}{\alpha'' + \beta''}$